

OFFCAMPUSCAREER PLATFORM

The Unified Corporate-Campus-Student Training, Internship, and Placement Ecosystem

A comprehensive product requirement, system design, and blueprint document detailing an end-to-end cloud platform for corporate hiring, AI-driven candidate screening, automated conversational interview conduct, campus course launching, and student skill-enhancement. Designed to bridge the corporate-academic divide and power the **Skilled India** initiative.

Document Title:	Offcampuscareer Implementation Proposal
Target Audience:	Corporate HRs, University TPOs, and Education Ministers
Version / Date:	v1.0.0 / June 2026 (Operational Draft)
Author Group:	Offcampuscareer Core Architectural Design Group
Document Status:	Approved / Ready for Development

Table of Contents & Executive Summary

Executive Summary & Table of Contents	Page 2
1. Vision & Alignment with 'Skilled India'	Page 3
2. Platform Overview & Ecosystem Architecture	Page 4
3. Corporate Portal - Job Posting & Multi-Channel Sourcing	Page 5
4. Corporate Portal - AI-Driven Candidate Screening & Scoring	Page 6
5. Corporate Portal - AI-Inbuilt Interview Conduct & Anti-Cheating	Page 7
6. Corporate Portal - AI Offer & Joining Letter Creation	Page 8
7. Campus Portal - Course Launching & Skill Development	Page 9
8. Campus Portal - Placement Drive Coordination	Page 10
9. Campus Portal - Executive Dashboard & Placement Analytics	Page 11
10. Student Portal - Course Enrollment & Learning Engine	Page 12
11. Student Portal - Job Board & Application Pipeline	Page 13
12. Student Portal - AI Career Prep & Practice Advisor	Page 14
13. Process Flowchart - Recruitment & Learning Journey	Page 15
14. System Architecture & Technology Stack	Page 16
15. Core Database Schema & Entities	Page 17
16. Market Analysis & Platform Projections	Page 18
17. Security, Compliance, and Data Privacy	Page 19
18. Product Roadmap & Gantt Timeline	Page 20

Executive Summary

The **Offcampuscareer** platform is conceptualized as an end-to-end cloud infrastructure designed to bridge the structural divide between corporations seeking skilled manpower and educational institutions looking to place students. The platform aggregates three primary hubs—Corporate, Campus, and Student—into a centralized marketplace.

By embedding advanced conversational AI, Automated Resume Screening, Remote Proctored Technical IDEs, and Smart Course builders, Offcampuscareer minimizes the recruitment overhead for enterprises while providing students with verifiable skill certifications and institutional backing. It addresses critical logistical constraints in placements and contributes directly to scaling a qualified workforce in India.

1. Vision & Alignment with 'Skilled India'

India possesses the largest demographic dividend in the world, with over 15 million youth entering the workforce annually. However, industry analysis consistently reveals an employability paradox: while millions of graduates are looking for jobs, corporations face severe shortages in skilled technical manpower. Traditional university curricula adapt slowly to rapid changes in industry requirements, leaving students equipped with outdated knowledge.

Furthermore, student career opportunities are highly fragmented. Top-tier companies heavily focus recruitment budgets on Tier-1 universities, leaving high-potential students in Tier-2 and Tier-3 cities with limited access to enterprise careers. Conversely, corporations miss out on broad talent pools due to the sheer logistical cost of traveling for physical placement drives or manually screening tens of thousands of online applications.

Core Strategic Goals

- **Democratizing Access:** Enabling students from any geographic location or college tier to compete fairly for top-tier internships and jobs based on objective, verified skill metrics.
- **Reducing Enterprise Time-to-Hire:** Accelerating corporate recruiting using advanced AI screening, reducing standard sourcing pipelines from 45 days down to less than 5 days.
- **Real-time Academic Alignment:** Empowering college administrators to instantly identify and deploy courses that match active corporate job descriptions, creating a tight feedback loop between skills and placement outcomes.
- **National Capacity Building:** Actively supporting the *Skilled India* initiative by training and certifying students on emerging technologies (AI, Cloud, Full Stack, Data Science) with corporate-endorsed certifications.

2. Platform Overview & Ecosystem Architecture

Offcampuscareer functions as a **unified multi-sided marketplace** that facilitates direct transactions and workflows between three critical stakeholder interfaces:

Stakeholder	Platform Interface Capabilities
Corporate Hub	Allows recruiters to post vacancies, set screening rules, configure AI conversational assessments, conduct automated live code compiler rounds, auto-generate offer/joining contracts, and track onboarding pipelines.
Campus Hub	Enables Training & Placement Officers (TPOs) and teachers to build and launch custom technical courses, upload video/quizzes, track student learning analytics, invite partners to campus placement drives, and view placement performance dashboards.
Student Hub	Provides candidates with custom learning paths, interactive coding playpens, virtual mock interview agents, resume grading tools, and a centralized application portal to apply for verified jobs and internships.

The heart of the platform is the **Central Matching Engine**. This engine processes data across the three pillars: it reads the courses completed by students, correlates their certified skill badges with active job criteria, and scores candidate fit. By structuring the ecosystem this way, we eliminate reliance on unverified resumes, moving instead to a verified-skills ledger where students showcase actual competencies backed by their academic institutions.

3. Corporate Portal - Job Posting & Multi-Channel Sourcing

For corporate recruiters, the platform replaces manual and fragmented sourcing with an **Enterprise-Grade Job Engine**. The process begins with dynamic job creation, where recruiters define more than just titles and descriptions:

- **Intelligent Job Specification Builder:** Instead of text boxes, recruiters define structured requirements, selecting required skill clusters, mandatory GPA/degree metrics, and candidate experience levels. The system uses these data points to build a semantic query profile.
- **Multi-Channel Placement Reach:** Once a recruiter publishes a job or internship, it instantly broadcasts to the dashboards of thousands of affiliated colleges and universities. Colleges can actively endorse the posting to specific cohorts, ensuring immediate, highly targeted applicant flow.
- **Recruiter Collaboration Workspace:** Large companies can assign recruiters and hiring managers to specific vacancies, manage access controls, leave feedback, and grade candidates concurrently in a joint kanban pipeline.
- **Custom Pre-Screening Assessments:** Companies can configure immediate gateway tests, choosing from a library of standard MCQ tests, behavioral question templates, or custom coding challenges that applicants must pass to proceed.

By organizing sourcing at the point of ingestion, recruiters avoid the need to sift through irrelevant resumes. The platform enforces data integrity from the start, requiring candidates to meet minimum eligibility thresholds defined by the employer before their application is finalized.

4. Corporate Portal - AI-Driven Candidate Screening

To handle massive volumes of applications without manual overhead, the platform integrates an **AI Candidate Screening and Resume Scoring engine**. Traditional screening relies on keyword matching, which often filters out qualified candidates who use different terminology. Offcampuscareer utilizes deep NLP (Natural Language Processing) and vector embeddings to perform semantic evaluations.

Key Capabilities of the Screening Engine

- **NLP Resume Extraction:** Automatically extracts contact details, education history, work experience, projects, and skills from uploaded PDF/Word resumes with over 98% accuracy.
- **Vector Skill Semantic Mapping:** Translates candidate details and job postings into multi-dimensional vectors. It matches conceptually related skills, meaning a search for 'NLP' will associate with candidates listing 'BERT, PyTorch, Transformers' even if 'NLP' is absent.
- **Dynamic Match Score:** Generates a normalized score (0-100) representing candidate suitability. This score weighs academic performance, course completion badges, mock interview performance, and resume relevance.
- **Candidate Ranking and Explainability:** Ranks the applicant pool instantly, providing hiring managers with clear summaries of why each candidate was scored high or low (e.g., 'Strong match in frontend framework, lacks SQL database experience').

This automated ranking helps companies identify the top 5% of candidates within minutes of a posting's release. Recruiting teams can focus on evaluating high-fit talent, significantly reducing administrative screening time and operational bottlenecks.

5. Corporate Portal - AI-Inbuilt Interview Conduct

Offcampuscareer features a fully integrated ****AI Conversational Interviewing Module****. This feature allows companies to scale the interview process without scheduling conflicts, conducting hundreds of technical and behavioral interviews simultaneously.

- **Conversational AI Interviewer:** An advanced, voice-enabled AI interviewer that conducts structural rounds. The AI reads the candidate's resume and dynamically generates custom questions, following up on their answers to probe depth of knowledge (e.g., asking how they solved a specific database bottleneck in their past project).
- **Real-time Technical IDE Compiler:** For software engineering roles, candidates are presented with a collaborative browser-based IDE supporting 20+ programming languages. The AI watches them code, assesses logic, time/space complexity, and tests it against edge cases.
- **Anti-Cheating and Identity Verification:** The system runs deep proctoring algorithms, including: multi-face detection, background voice detection, browser-tab switching locks, copy-paste disables, and automated webcam recording with flags for suspicious behavior.
- **Comprehensive Scorecards:** Post-interview, the AI compiles a detailed performance scorecard grading communication, problem-solving speed, technical correctness, and logical consistency, with an audio-video recording for human review.

This conversational approach provides recruiters with rich data points that standard multiple-choice tests fail to deliver. The conversational AI can assess behavioral fit, situational responses, and problem-solving structures, ensuring high-quality hiring signals.

6. Corporate Portal - AI Offer & Joining Letter Creation

The final mile of the hiring process is typically slowed down by administrative friction. Offcampuscareer addresses this with its ****AI Contract Generation Engine****, which automates document compilation, legal review, and signature collection:

- **Dynamic Contract Assembler:** The AI reads the candidate's profile, job details, and approved compensation structure, automatically assembling a personalized offer letter using templates customized for different roles and regions.
- **AI Compliance Auditor:** Prior to issuance, the system checks the drafted agreement against state employment regulations, verifying clauses for notice periods, benefits, and NDAs, ensuring compliance with local laws.
- **Integrated E-Signature Engine:** Offers a secure, audit-trailed digital signature flow (compliant with Information Technology Act standards). Recruiters sign and send the document, and candidates accept and sign it directly on their dashboard.
- **Automated Joining Instructions:** Once the offer is accepted, the system instantly generates and emails a custom Joining Letter. This document details office locations, required documentation (ID, educational credentials), onboarding schedules, and IT setups.

By integrating contract generation directly with candidate records, HR operations become seamless. The onboarding pipeline shifts from manual email threads to a single dashboard where signing status, background checks, and orientation events are tracked in real time.

7. Campus Portal - Course Launching & Skill Development

For academic institutions, the platform functions as an ****Employability-Focused Learning Management System (LMS)****. College administrators, Training & Placement Officers (TPOs), and professors can create and launch specialized courses to address industry skill gaps:

- **Curriculum & Syllabus Builder:** Professors can quickly structure courses, upload lecture videos, import documents, and create assessments (quizzes, coding exercises). The interface supports rich media and external link integration.
- **Verified Skill Mapping:** Every course module is associated with industry skills. When a student finishes a course and passes the final exam, the platform awards a verified 'Skill Badge' directly to their student profile.
- **Corporate Co-Authored Pathways:** Universities can co-author courses with recruiting companies, utilizing content directly curated by corporate partners. Students completing these courses are marked as priority candidates for that partner's vacancies.
- **LMS Analytics Integrations:** The system syncs with existing university database tools, importing rosters and exporting grades, reducing administrative double-entry.

This educational framework shifts the academic focus from generic theory to market-driven skills. By publishing courses aligned with corporate needs, universities can actively raise their placement statistics, ensuring students graduate with credentials that recruiters trust.

8. Campus Portal - Placement Drive Coordination

Traditional campus placement drives are logistical nightmares, requiring manual Excel tracking, scheduling across multiple rooms, and endless email chains. Offcampuscareer introduces the ****Unified Placement Drive Coordinator****:

- **Virtual Placement Event Manager:** Campuses can create placement events (e.g., 'Annual Tech Internship Drive'), set dates, and send invitations directly to registered companies. Corporations review the event and RSVP online.
- **Smart Scheduling Coordinator:** System automatically organizes interview schedules, matching student availabilities with corporate interviewer slots. It handles round-by-round progressions (e.g., automatically promoting candidates to subsequent rounds upon interviewer approval).
- **Digital Resume Book Builder:** TPOs can generate and export standardized resume books containing student profiles, verified skills, and academic transcripts, allowing recruiters to review cohorts prior to the drive.
- **Real-Time Offer Tracking:** TPOs can view selections in real time as recruiters log decisions. This tracking eliminates communication delays and helps colleges track their placement target metrics.

Digitizing drives makes campus recruitment scalable. TPOs spend less time managing calendars and spreadsheets and more time building corporate partnerships and advising students.

9. Campus Portal - Executive Dashboard & Analytics

Academic leaders and TPOs require clear data to make strategic decisions about curriculum enhancements and student preparation. The platform provides an ****Executive Analytics Suite**** that visualizes student progress and institutional results:

- **Curriculum & Course Performance:** Monitors student enrollment, drop-off points, and final grades across launched courses. Administrators can identify which courses are highly engaging and which require better instructional support.
- **Skills Heatmap:** Aggregates student profiles to visualize the university's technical strength (e.g., showing 60% of engineering students are proficient in React, but only 10% in SQL), helping TPOs pitch specific talent pools to matching companies.
- **Placement Metrics Dashboard:** Displays placed vs. unplaced student ratios, average package offerings, and top-recruiting companies. This data can be filtered by branch, degree, or graduation year.
- **Employer Feedback Aggregator:** Collects post-interview and post-internship feedback from corporate partners, identifying common skill gaps in the student body and recommending curriculum updates.

With access to aggregated dashboard analytics, universities can move away from speculative academic planning. They can base their educational offerings on direct corporate demand, increasing placement rates and student success.

10. Student Portal - Learning Platform & Certification

For students, the platform is a comprehensive learning and career engine. When students log in, they are presented with a ****Personalized Learning Environment**** tailored to their career aspirations:

- **Personalized Career Maps:** Students select their target roles (e.g., Backend Developer, Cybersecurity Specialist). The platform then outlines a tailored roadmap detailing the courses they must complete and badges they must earn.
- **Interactive Coding Environments:** Integrates web-based sandboxes for JavaScript, Python, and SQL. Students write and test code directly in the browser, completing practical labs and assignments without local environments.
- **Self-Paced Learning Engine:** Tracks video progress, bookmarks resources, and supports student discussion forums. Students can ask questions and collaborate on course topics with peers and instructors.
- **Immutable Skill Verification:** Course completion is validated through structured exams. Earning a badge updates the student's profile, making it immediately visible to hiring recruiters searching the platform.

By linking learning directly to career placement, students are motivated to finish courses. Their efforts result in verifiable credentials, giving them an advantage when applying for jobs and internships on the platform.

11. Student Portal - Job Board & Application Pipeline

The ****Student Job Application Engine**** removes the confusion from searching for jobs and internships. The portal focuses on clarity, transparency, and one-click applications:

- **Smart Skill-Matched Job Board:** Rather than sorting through keywords, the dashboard highlights opportunities matching the student's verified skill badges and GPA, displaying compatibility percentages (e.g., '95% skill match').
- **Unified Candidate Profile:** Instead of uploading separate resumes, students share their verified profile, which includes their academic history, course certificates, coding scores, and verified skill badges.
- **Kanban Application Tracker:** Tracks application states in real time. Students see exactly where they stand in the hiring process: Applied, Under Review, Interviewing, Offered, or Closed.
- **Direct Recruiter Chat:** Offers a built-in messaging channel for recruiters and candidates. Recruiters can request additional documents, schedule follow-ups, and chat with applicants directly.

This structured application process builds trust. Students are no longer applying to black-hole job boards; they have clear visibility into their status, receive direct feedback, and access roles backed by institutional support.

12. Student Portal - AI Career Advisory and Preparation

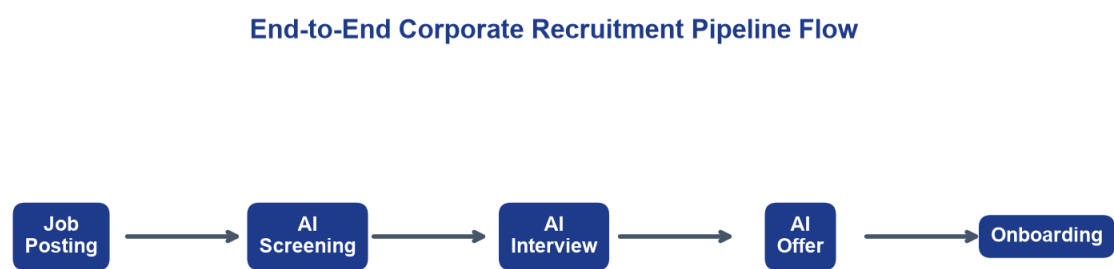
Students in Tier-2 and Tier-3 campuses often lack access to professional placement advisors. Offcampuscareer addresses this by providing every student with a ****Personalized AI Career Coach****:

- **AI Resume Optimizer:** The AI analyzes student resumes, flagging formatting issues, highlighting missing keywords, and recommending structural changes to align with their target job descriptions.
- **Unlimited AI Mock Interviews:** Students can practice behavioral and technical mock interviews with the conversational AI agent. The AI asks role-specific questions and provides feedback on communication, coding structure, and content.
- **Career Gap Identifier:** Analyzes the student's profile against active market vacancies, identifying missing technical skills and suggesting specific courses to close the gap.
- **Assessment Practice Portal:** A library of mock MCQs and coding challenges, helping students build confidence and practice for corporate entrance tests.

This continuous feedback loop allows students to improve their skills before interviewing. The AI coach acts as an accessible mentor, ensuring students are prepared for competitive corporate assessments.

13. Process Flowchart

The diagram below illustrates the end-to-end recruitment process. It maps how a company's job posting connects with a student's educational accomplishments, triggering AI screening, automated conversational interviews, contract generation, and onboarding:



Workflow Phase Breakdown:

- 1. **Job Posting & Distribution:** Recruiters post roles and define required skill badges, which broadcast to affiliated campus dashboards.
- 2. **AI Sourcing & Screening:** Students apply with verified profiles. The AI screening engine matches their credentials, generating fit scores.
- 3. **AI Conversational Evaluation:** High-scoring candidates enter the AI interview portal for automated coding and behavioral rounds.
- 4. **Contract Generation:** AI uses approved templates and candidate records to draft, audit, and send offer letters.
- 5. **Digital Onboarding:** Candidates accept and sign letters online, triggering automated joining instructions.

14. System Architecture & Technology Stack

To ensure scalability, security, and high availability, Offcampuscareer is designed as a ****Microservices Cloud Architecture**** hosted on Google Cloud Platform. The system decouples high-throughput transactional flows from computational AI tasks.

Layer	Technologies Used	Architectural Role
Frontend Portal	React.js, TypeScript, Tailwind CSS, shadcn/ui	Responsive, interactive client interfaces for Students, Recruiters, and TPOs.
Application APIs	Node.js (Express), Go, REST, WebSocket	Core business rules, placement drives, chats, and notification alerts.
AI Services Engine	Python (FastAPI), LangChain, PyTorch	Resume parsing, mock interviews, dynamic test grading, and contract audits.
Databases	PostgreSQL, MongoDB, Redis Cache	Postgres for transactions, MongoDB for course content/resumes, Redis for user sessions.
LLM Model API	Google Gemini 1.5 Pro & Flash APIs	Generates interview questions, scores responses, and draft agreements.
Infrastructure	Docker, Kubernetes, GKE, Google Cloud	Containerized deployments, auto-scaling, and secure file storage in GCS.

API requests are routed through an API Gateway that manages rate limiting, authentication, and logging. WebSockets maintain real-time updates for chat and placement drive boards. The AI pipeline runs asynchronously using task queues, ensuring stable frontend performance during heavy workloads.

15. Core Database Schema & Entities

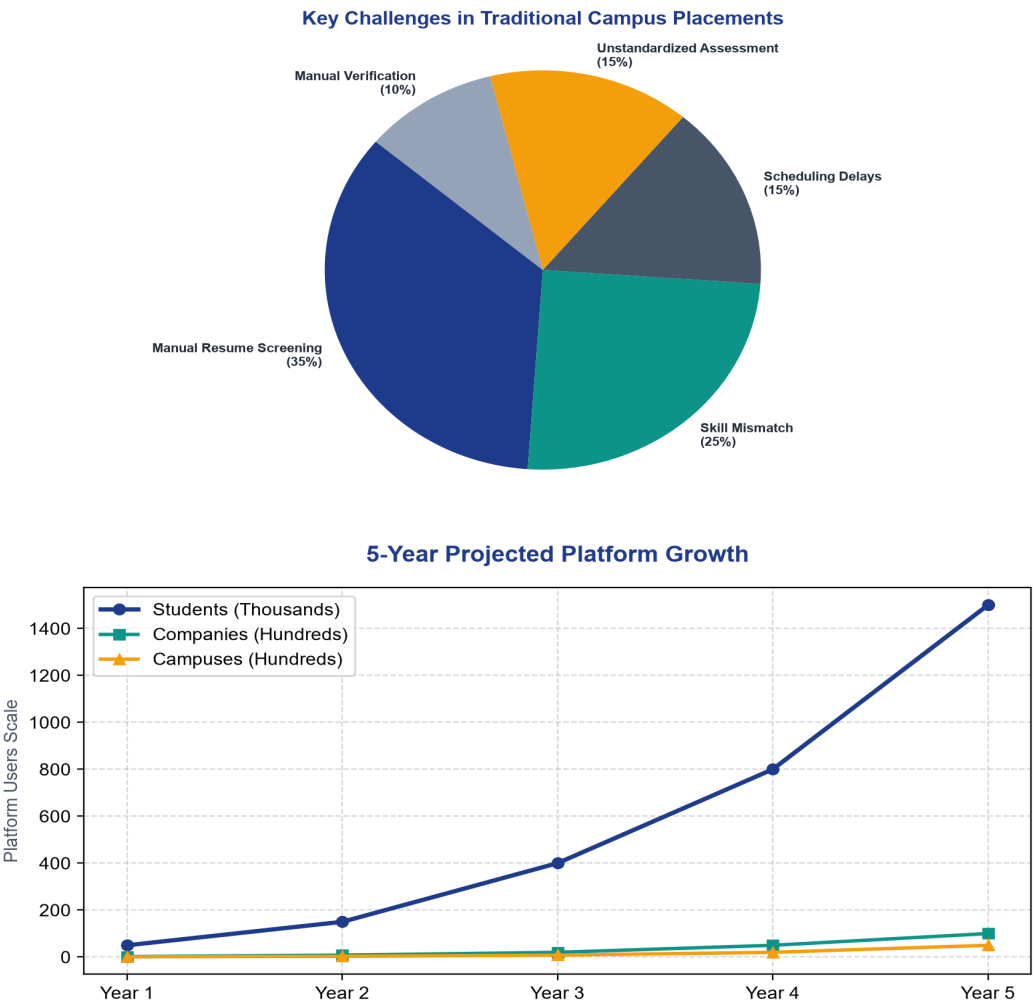
The platform's data model is structured around a relational schema for transactions, supplemented by unstructured document stores for course syllabi and resume data. The table below outlines our core database entities and relationships:

Entity Name	Primary Keys & Fields	Relationships & Constraints
Users	id (UUID), email, password_hash, role (Enum: Student, Recruiter, TPO, Admin), created_at	One-to-many with JobPostings (Recruiters) and Applications (Students).
JobPostings	id (UUID), company_id, title, description, skills_required (JSON), min_gpa, salary, status	Belongs to a Recruiter. One-to-many with Applications.
Applications	id (UUID), job_id, student_id, resume_url, status, match_score, submitted_at	Foreign keys to Jobs and Students. One-to-one with OfferLetters.
Interviews	id (UUID), application_id, scheduled_at, ai_video_url, evaluation_score, feedback_report	Belongs to an Application. References candidate assessment records.
Courses	id (UUID), campus_id, title, syllabus (JSON), enrolled_count, badge_id (UUID)	Belongs to a Campus (TPO). Many-to-many with Students via enrollments.
OfferLetters	id (UUID), application_id, contract_text, status (Draft, Sent, Signed), signed_at	Belongs to an Application. Enforces e-signature hash keys.

PostgreSQL enforces referential integrity for core applicant status changes, interview scores, and offer acceptances. Resume files and course video metadata are managed securely in Google Cloud Storage, with public-key signed URLs restricting access to authorized users.

16. Market Analysis & Platform Projections

Our research reveals a significant market gap in the entry-level recruitment ecosystem. Standard job portals are optimized for experienced hires and struggle to serve entry-level candidates, who lack verified histories:



As illustrated, 35% of recruitment delays stem from manual resume screening. By replacing this phase with automated AI matching and institutional skill verifications, Offcampuscareer saves considerable resource overhead. Our 5-year growth plan targets onboarding 1.5 million active students, 1,000 partner companies, and 500 academic campuses, driving network density and standardizing placement pipelines.

17. Security, Compliance, and Data Privacy

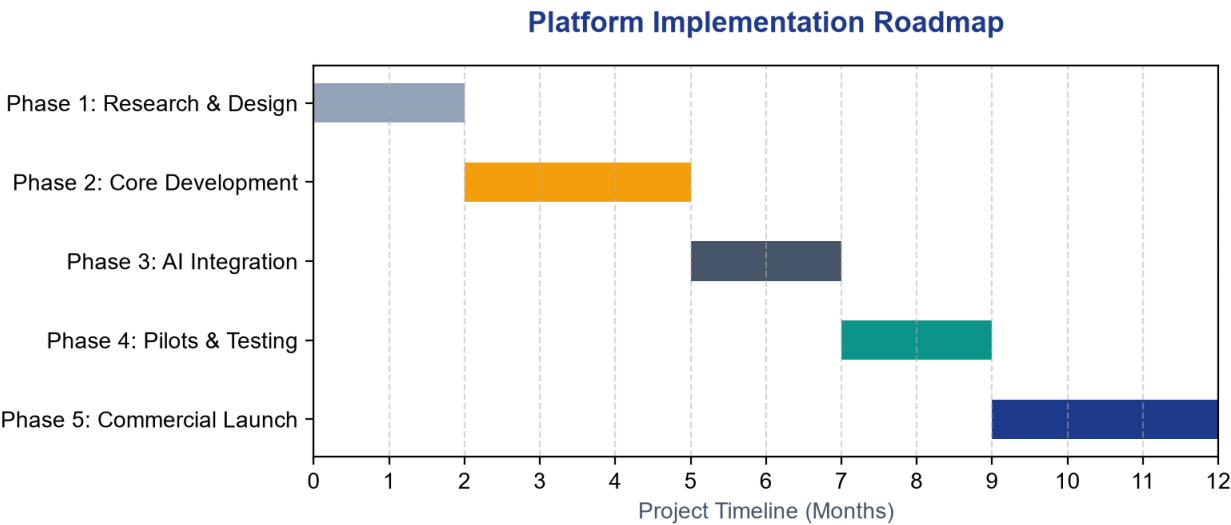
Handling student transcripts, resume records, corporate recruitment criteria, and webcam proctoring feeds requires high standards of security and privacy:

- **Data Encryption:** All data is encrypted in transit using TLS 1.3 and at rest in our databases and GCS buckets using AES-256 keys, protecting sensitive candidate profiles from unauthorized access.
- **Tenant Isolation:** Logical firewalls and access control layers prevent cross-tenant data leakage. College administrators can only access rosters from their own campus, and recruiters can only view details of their applicants.
- **GDPR and DPDP Act Alignment:** In compliance with India's Digital Personal Data Protection Act, students retain ownership of their data. They can request to delete their profile, revoke sharing permissions, or audit active recruiters.
- **AI Proctoring Privacy Guard:** Video and audio streams captured during proctored exams are processed via sandboxed containers. This data is only used to generate anti-cheat scorecards and is deleted after 30 days.
- **Compliance Roadmap:** The platform's infrastructure is built to meet SOC 2 Type II and ISO 27001 requirements, preparing it for enterprise deployment with security compliance audits.

Role-Based Access Control (RBAC) ensures administrative operations require multi-factor authentication. In addition, the platform uses automated monitoring to detect SQL injections, cross-site scripting (XSS), and DDoS attacks, protecting system availability.

18. Product Roadmap & Gantt Timeline

The timeline below details our 12-month roadmap to build, integrate, test, and launch the Offcampuscareer platform. We begin with user research and core architecture, followed by AI development and pilot campus rollouts:



Key Delivery Milestones:

- **Months 1-2 (Phase 1):** Draft product requirements, system design schemas, database structures, and responsive UI wireframes.
- **Months 3-5 (Phase 2):** Build core database engines, user registries, the job board, and course learning dashboards.
- **Months 6-7 (Phase 3):** Integrate AI services (resume parser, conversational mock interviews, e-signature generator).
- **Months 8-9 (Phase 4):** Conduct beta tests and run placement pilots with 10 selected campuses and 50 corporate partners.
- **Months 10-12 (Phase 5):** Deploy production environments, launch corporate outreach campaigns, and open student registration.

Conclusion: By integrating training, internships, and placement coordination into a single ecosystem, Offcampuscareer offers a modern solution to career development. By deploying automated AI tools, the platform reduces administrative overhead, helping academic institutions build a skilled workforce and assisting corporations in sourcing talent efficiently.